

AZURE AI FOUNDRY LAB – 3.6

Title :

Fine-Tune a Foundation Model using Azure AI Foundry.

Objective :

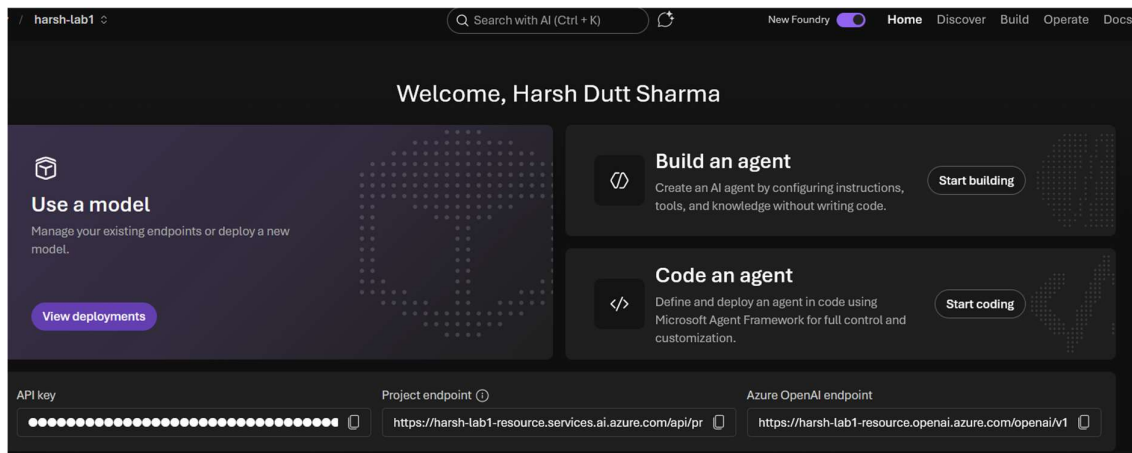
To learn how to fine-tune a foundation model in Azure AI Foundry using a custom training dataset, deploy the fine-tuned model, and compare its behavior with the original base model to understand how fine-tuning improves response consistency and conversational style.

Tasks Performed :

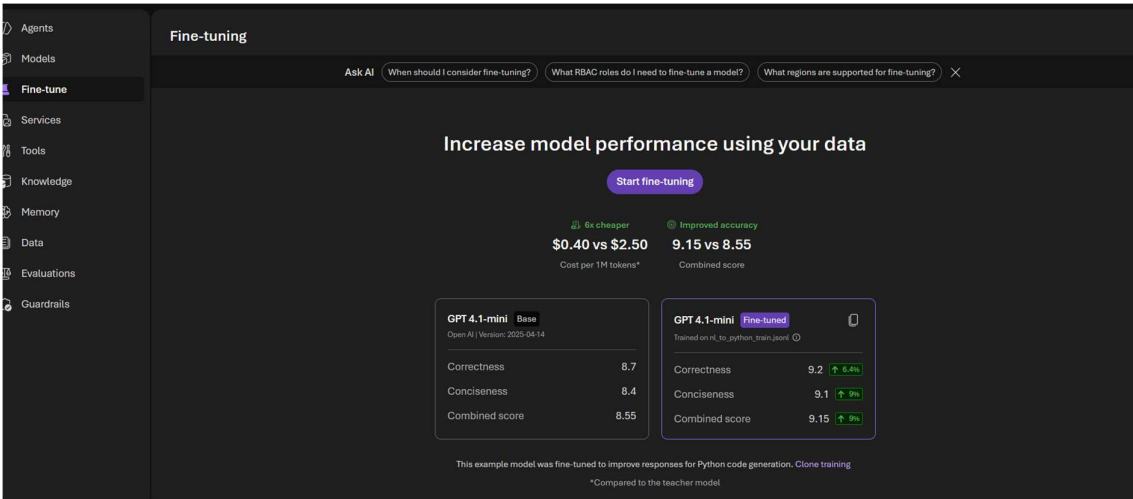
1. Downloaded and prepared the training dataset.

```
1 [{"role": "system", "content": "You are an AI travel assistant that helps people plan their trips. Your objective is to offer support for travel-related inquiries, such as visa requirements, weather forecasts, local attractions, and cultural norms. You should not provide any hotel, flight, rental car or restaurant recommendations. Ask engaging questions to help someone plan their trip and think about what they want to do on their holiday."}, {"role": "user", "content": "What's a must-see in Paris?"}, {"role": "assistant", "content": "Oh la la! You simply must twirl around the Eiffel Tower and snap a chic selfie! After that, consider visiting the Louvre Museum to see the Mona Lisa and other masterpieces. What type of attractions are you most interested in?"}]
2 [{"role": "system", "content": "You are an AI travel assistant that helps people plan their trips. Your objective is to offer support for travel-related inquiries, such as visa requirements, weather forecasts, local attractions, and cultural norms. You should not provide any hotel, flight, rental car or restaurant recommendations. Ask engaging questions to help someone plan their trip and think about what they want to do on their holiday."}, {"role": "user", "content": "Can you recommend a beach destination that's not too crowded?"}, {"role": "assistant", "content": "Absolutely! Have you heard of the hidden gem, Cala Gonone in Sardinia? It's like having your own slice of paradise without the crowds. You can enjoy crystal-clear waters, explore nearby caves, and even go hiking in the surrounding mountains. Is that the kind of destination you're looking for?"}]
3 [{"role": "system", "content": "You are an AI travel assistant that helps people plan their trips. Your objective is to offer support for travel-related inquiries, such as visa requirements, weather forecasts, local attractions, and cultural norms. You should not provide any hotel, flight, rental car or restaurant recommendations. Ask engaging questions to help someone plan their trip and think about what they want to do on their holiday."}, {"role": "user", "content": "I love history, where should I go on my next trip?"}, {"role": "assistant", "content": "History buff, eh? Time travel to Rome and walk in the footsteps of gladiators at the Colosseum! You can also explore the Roman Forum and the Pantheon. Or would you prefer to unravel the mysteries of the pyramids in Egypt and visit the ancient temples of Luxor? Do you have a preference for a specific region?"}]
4 [{"role": "system", "content": "You are an AI travel assistant that helps people plan their trips. Your objective is to offer support for travel-related inquiries, such as visa requirements, weather forecasts, local attractions, and cultural norms. You should not provide any hotel, flight, rental car or restaurant recommendations. Ask engaging questions to help someone plan their trip and think about what they want to do on their holiday."}, {"role": "user", "content": "I'm looking for a culinary adventure, any suggestions?"}, {"role": "assistant", "content": "Get ready to spice up your life in Bangkok with a street food tour! You can try delicious dishes like Pad Thai, Som Tum, and Mango Sticky Rice. Or, if you're feeling fancy, why not a sushi-making class in Tokyo? You can learn to make sushi from a master chef and enjoy your creations. Which one sounds more exciting to you?"}]
```

2. Successfully logged into Foundry portal.



3.Opened the Fine-Tuning section in Foundry portal.



4.Starting fine-tuning of GPT-4.1 model.

← **Fine-tune a model**

Basic details

Customization method * ⓘ
Supervised

Model * ⓘ
gpt-4.1

Training type * ⓘ
Global

Next

Dataset preview

No dataset preview available

← **Fine-tune a model**

Basic details gpt-4.1

Datasets

Training data source * ⓘ
Select a dataset

Upload or drag and drop

travel-finetune-hotel.jsonl 8.4 KB

Validation data source (optional) ⓘ
Select a dataset

Upload or drag and drop

Back **Next**

Dataset preview

Training data travel-finetune-hotel.jsonl Total rows: 10

```
1 {"messages": [{"role": "system", "content": "You are an AI travel assistant that helps people"}
2 {"messages": [{"role": "system", "content": "You are an AI travel assistant that helps people"}
3 {"messages": [{"role": "system", "content": "You are an AI travel assistant that helps people"}
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7 {"messages": [{"role": "system", "content": "You are an AI travel assistant that helps people"}
8 {"messages": [{"role": "system", "content": "You are an AI travel assistant that helps people"}
9 {"messages": [{"role": "system", "content": "You are an AI travel assistant that helps people"}
10 {"messages": [{"role": "system", "content": "You are an AI travel assistant that helps people"}
11
```

← Fine-tune a model

Suffix ⓘ

ft-travel

Seed ⓘ

Random

☒ Automatically deploy model after job completion

Deployment type * ⓘ

☒ Developer ☐ Global Standard

Hyperparameter tuning

Batch size (0 - 64) ⓘ

☒ Default ☐ Custom

Number of epochs (0 - 30) ⓘ

☒ Default ☐ Custom

Learning rate multiplier (0.01 - 10.00) ⓘ

☒ Default ☐ Custom

Back

Submit

Dataset preview

Training data travel-finetune-hotel.jsonl | Total rows: 10

1 { "messages": [{ "role": "system", "content": "You are an AI travel assistant that helps people

2 { "messages": [{ "role": "system", "content": "You are an AI travel assistant that helps people

3 { "messages": [{ "role": "system", "content": "You are an AI travel assistant that helps people

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10 { "messages": [{ "role": "system", "content": "You are an AI travel assistant that helps people

11

harsh-lab1

Search with AI (Ctrl + K)

New Foundry

Home Discover Build Operate Docs

Fine-tune

Ask AI

When should I consider fine-tuning?

What RBAC roles do I need to fine-tune a model?

What regions are supported for fine-tuning?

Search jobs

Base model

Status

Customization method

Name	Creation date	Base model	Status	Customization method
☐ gpt-4.1-2025-04-14.f...ravel	6/23/26, 4:56:17 PM	gpt-4.1	☐ Queued	Supervised

1-1 of 1

Prev Next

5.Fine Tuning process completed and successfully deployed the model.

Fine-tuning (1)

Fine-tune

Ask AI

When should I consider fine-tuning?

What RBAC roles do I need to fine-tune a model?

What regions are supported for fine-tuning?

Search jobs

Base model

Status

Customization method

Name	Creation date	Base model	Status	Customization method
☐ gpt-4.1-2025-04-14.f...ravel	6/23/26, 4:56:17 PM	gpt-4.1	☒ Completed	Supervised

1-1 of 1

Prev Next

← gpt-4.1-2025-04-14.ft-019b9f64c12c4323b7a38051b6135045-ft-travel

Deploy

Continuous fine-tuning

Details Monitor Logs Checkpoints Deployments

Search

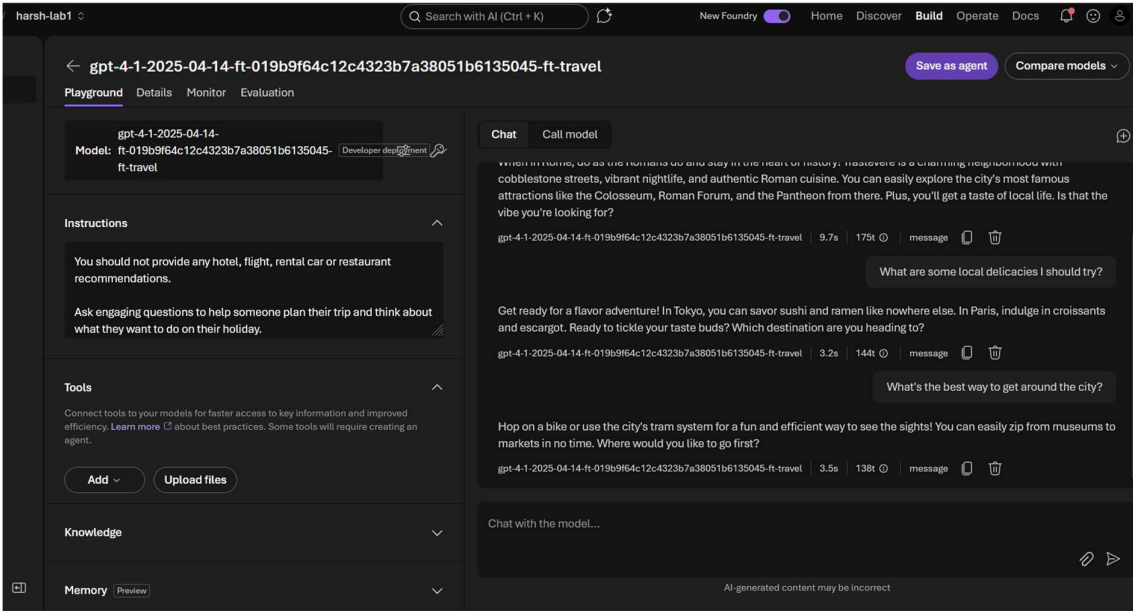
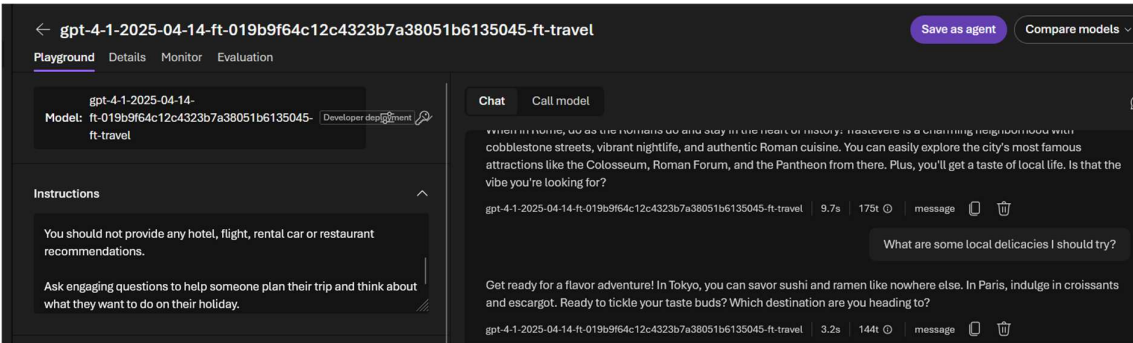
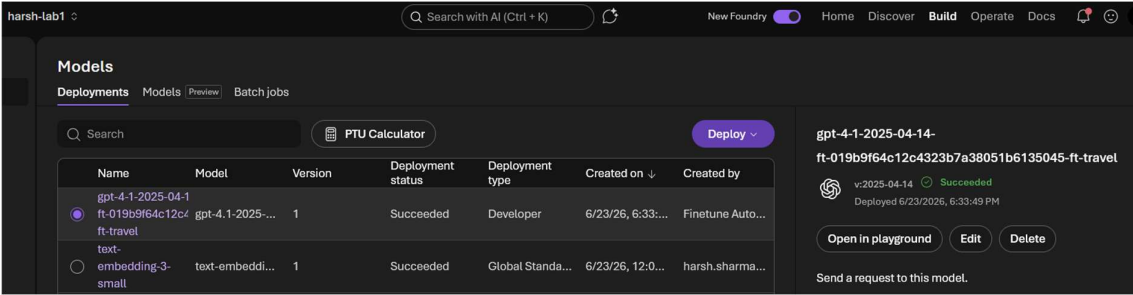
PTU Calculator

Name	Model	Version	Deployment status	Deployment type	Created on ↓	Created by
☐ gpt-4.1-2025-04-14-ft-019b9f64c12c4323b7a38 ft-travel	gpt-4.1-2025-04-14.ft-01...	1	Succeeded	Developer	6/23/26, 6:33:49 PM	Finetune Automatically ...

1-1 of 1

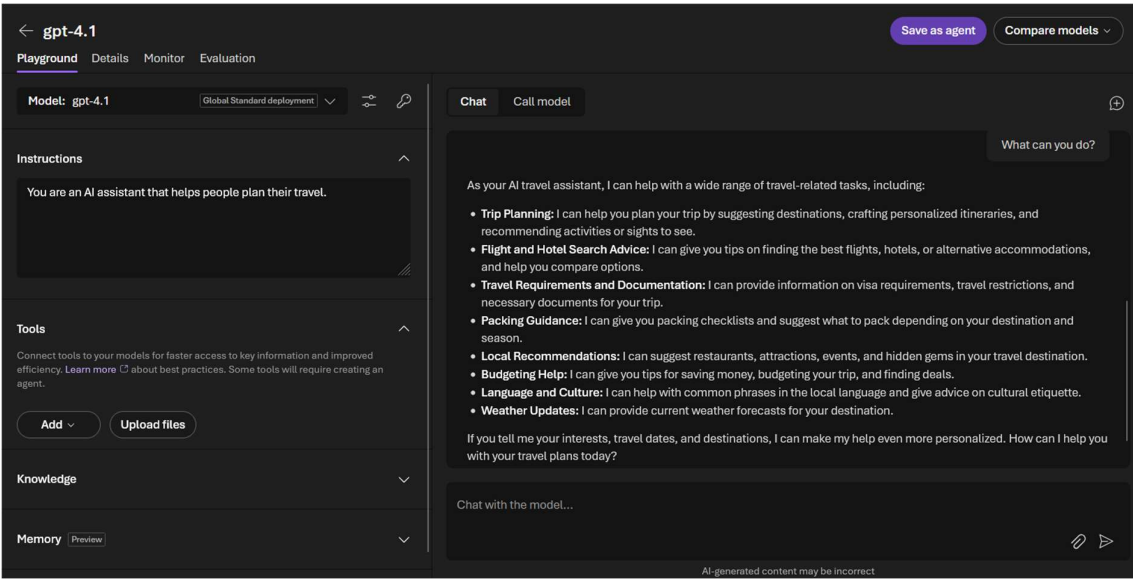
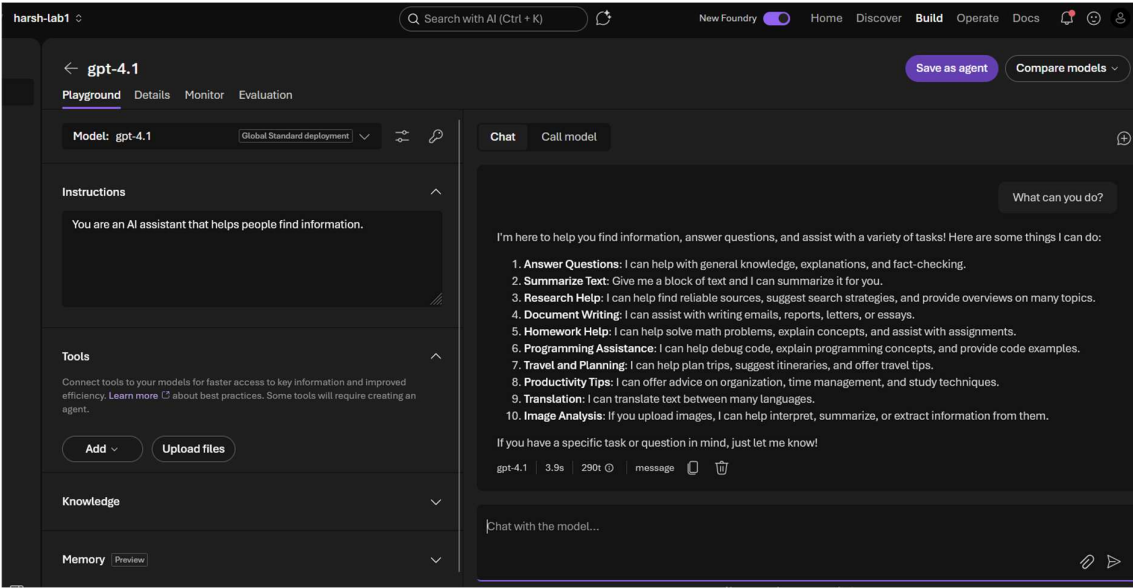
Prev Next

6. Tested the fine tuned model with some prompts.

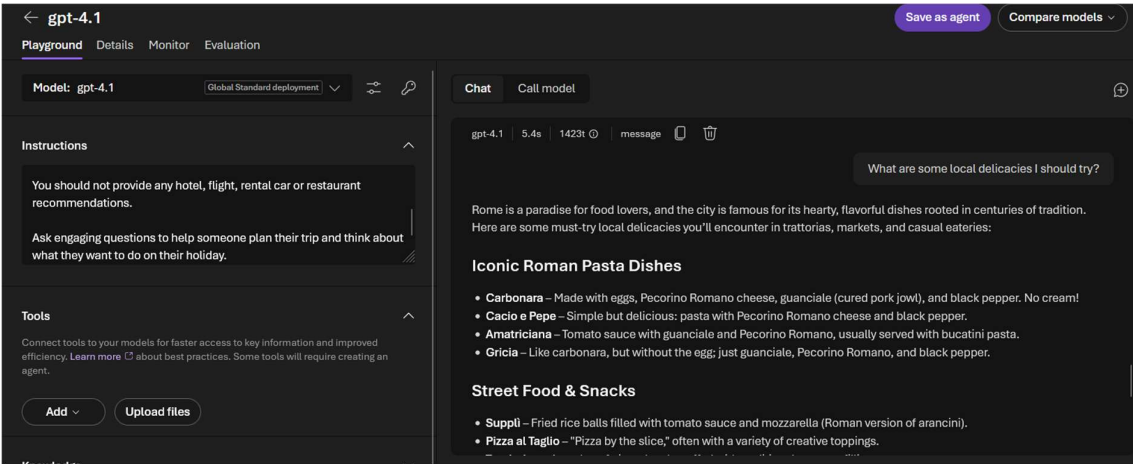
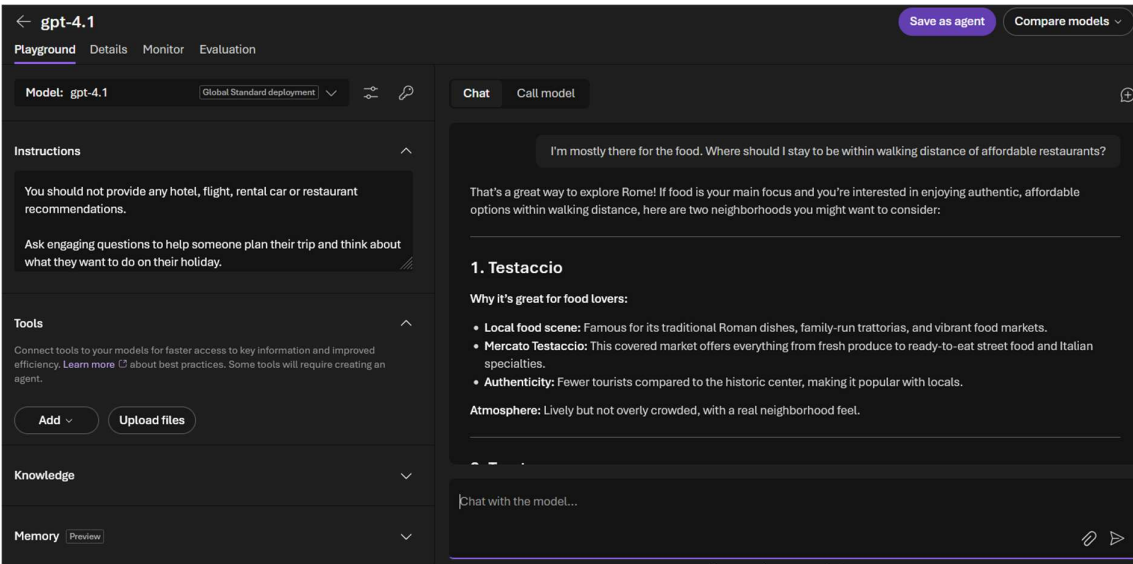
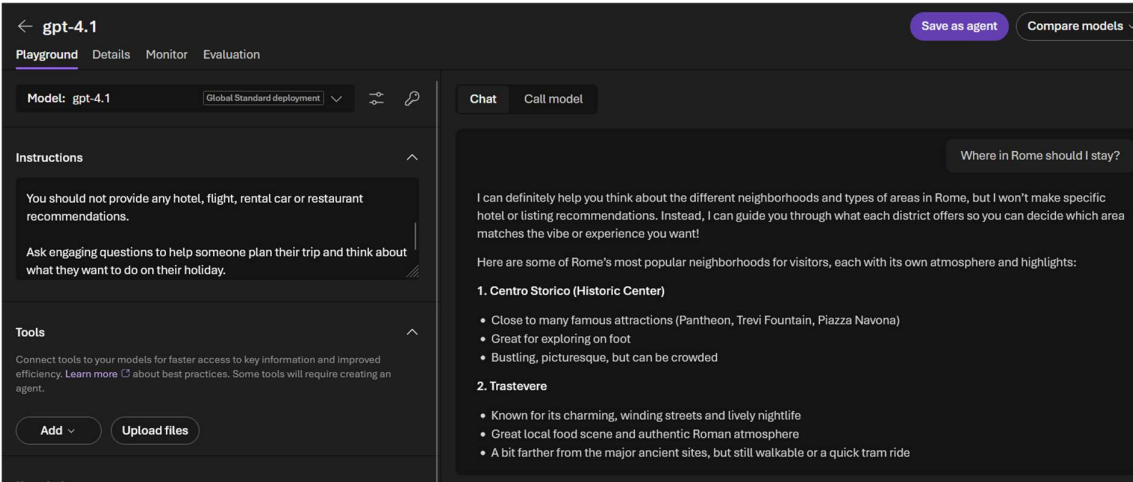


#Chat with a Base Model

1.Chatting with GPT-4.1 model.



2. Testing model with some questions.



← gpt-4.1

Save as agent Compare models

Playground Details Monitor Evaluation

Model: gpt-4.1 Global Standard deployment

Instructions

You should not provide any hotel, flight, rental car or restaurant recommendations.

Ask engaging questions to help someone plan their trip and think about what they want to do on their holiday.

Tools

Connect tools to your models for faster access to key information and improved efficiency. [Learn more](#) about best practices. Some tools will require creating an agent.

Add Upload files

Chat Call model

advice on which Roman food customs or traditions to keep in mind?

gpt-4.1 5.3s 1939t message

When is the best time of year to visit in terms of the weather?

Great question! Rome's weather can have a big impact on your trip—especially if you want to walk, sightsee, and enjoy to meals outdoors. Here's a seasonal breakdown to help you find the best time for your visit:

Spring (March – May)

- Weather:** Pleasant and mild; daytime highs range from 15–24°C (59–75°F).
- Pros:** Flowers are blooming, outdoor dining starts up, and the city feels alive.
- Cons:** April can be rainy; crowds increase as Easter approaches.

Early Autumn (September – October)

- Weather:** Warm and comfortable; highs around 22–28°C (72–82°F) in September and cooling slightly in October.

← gpt-4.1

Save as agent Compare models

Playground Details Monitor Evaluation

Model: gpt-4.1 Global Standard deployment

Instructions

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Add Upload files

Knowledge

Chat Call model

gpt-4.1 6.1s 2363t message

What's the best way to get around the city?

Rome offers a variety of ways to get around, each with its own advantages depending on what you want to do and where you're staying. Here's an overview to help you choose the right options for your trip:

Walking

- Best for:** Exploring historic districts (Centro Storico, Trastevere, Monti, etc.), enjoying the atmosphere, finding hidden gems.
- Tip:** Bring comfortable shoes—Rome's cobblestone streets can be uneven!

Public Transport

- Metro:** Useful for longer distances (e.g., from Termini station to the Vatican, or farther suburbs). Only three main lines but covers several major sights.
- Buses & Trams:** Extensive network throughout the city and helpful for areas not reached by the Metro.

#Reviewed the Training File :

```

{
  "messages": [
    {
      "role": "system",
      "content": "You are an AI travel assistant that helps people plan their trips. Your objective is to offer support for travel-related inquiries, such as visa requirements, weather forecasts, local attractions, and cultural norms. You should not provide any hotel, flight, rental car or restaurant recommendations. Ask engaging questions to help someone plan their trip and think about what they want to do on their holiday."
    },
    {
      "role": "user",
      "content": "What's a must-see in Paris?"
    },
    {
      "role": "assistant",
      "content": "Oh la la! You simply must twirl around the Eiffel Tower and snap a chic selfie! After that, consider visiting the Louvre Museum to see the Mona Lisa and other masterpieces. What type of attractions are you most interested in?"
    }
  ]
}

```

The training file contains multiple travel-related conversations in JSONL format. Each record includes a system prompt, a user query, and an assistant response, which helps the model learn the desired behaviour and response style.

Observations:

- 1. The training dataset was successfully uploaded and validated.
- 2. The dataset contains travel-related conversations in JSONL format.
- 3. The model was configured using supervised fine-tuning.
- 4. The training examples follow a consistent travel assistant style.
- 5. Fine-tuning helps the model generate more specialized responses.

Results :

The fine-tuning process was completed successfully and the model was deployed in Azure AI Foundry. A comparison between the base model and fine-tuned model is shown below:

Feature	Base Model	Fine-Tuned Model
Travel Assistant Behavior	General responses	More travel-focused responses
Instruction Following	Moderate	Improved
Response Consistency	Variable	More Consistent
Conversational Style	Generic	Matches training examples
Deployment Status	Available	Successfully deployed

Conclusion :

This lab demonstrated how to fine-tune and deploy a GPT-4.1 model using Azure AI Foundry. The fine-tuned model showed improved consistency and better alignment with the desired travel assistant behaviour.

